

# APEX INSTITUTE OF TECHNOLOGY & SCIENCES (AUTONOMOUS)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

## CONTINUOUS TEST 2 (CT-2) EXAMINATION • AY 2025-26

Course: Database Management Systems (CS501) | Faculty: Dr. Ravi Kumar

Date: 28 October 2025

Time: 60 Minutes

Max Marks: 20

Instructions: Answer all questions in Part A. Part B contains internal choice.

### PART A — THEORY OF RELATIONAL DECOMPOSITION

Answer both questions.

1. State Armstrong Axioms for functional dependencies (Transitivity, Augmentation, Reflexivity). [4 M]
2. Explain the difference between 3NF and BCNF with a counterexample. [6 M]

### PART B — NORMALIZATION & SQL QUERIES

Solve the relational problem.

3. Given relation  $R(A, B, C, D, E)$  with  $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ . Find all candidate keys and decompose  $R$  into BCNF. [10 M]